The gradient of the function is

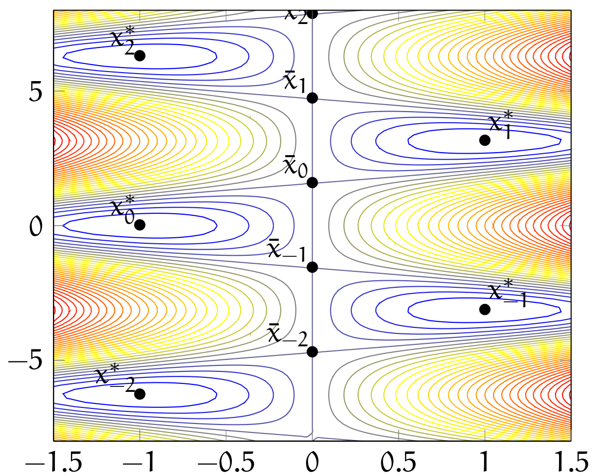
$$\nabla f(x_1, x_2) = \begin{pmatrix} x_1 + \cos x_2 \\ -x_1 \sin x_2 \end{pmatrix}.$$

This gradient is zero for

$$x_k^* = \begin{pmatrix} (-1)^{k+1} \\ k\pi \end{pmatrix}, \; k \in \mathbb{Z},$$

and for

$$\bar{x}_k = \begin{pmatrix} 0 \\ \frac{\pi}{2} + k\pi \end{pmatrix}, \; k \in \mathbb{Z}.$$



The second derivative matrix is

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} 1 & -\sin x_2 \\ -\sin x_2 & -x_1 \cos x_2 \end{pmatrix}.$$

By evaluating this matrix at x_k^*, we obtain for any $k \in \mathbb{Z}$

$$\nabla^2 f(x_k^*) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since this matrix is positive definite, each point x_k^* satisfies the sufficient optimality conditions and is a local minimum of the function.

By evaluating the second derivative matrix at $\bar{x}_k$, we obtain for any $k \in \mathbb{Z}$

$$\nabla^2 f(\bar{x}_k) = \begin{pmatrix} 1 & (-1)^{k+1} \\ (-1)^{k+1} & 0 \end{pmatrix}.$$

Regardless of k, this matrix is not positive semidefinite. Indeed, if k is even,

$$\nabla^2 f(x_k^*) = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}.$$

If k is odd,

$$\nabla^2 f(x_k^*) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

In both cases, the eigenvalues are -0.61803 and 1.61803.

Therefore, there is no $\bar{x}_k$ that satisfies the necessary optimality conditions. None of them can then be a local minimum.